

1. Online Process



Capture Device

On-device Real-time Detection & Control



Hand Detect

Auto recording
Voice remind



Operate Specs

Gesture detect
Scene detect



Wear Specs

Wearing check
Stability check



Light Environment

Low-light detect
Auto-exposure



Real-time Performance

Deblur, Adaptive
frame rate, Auto stop

Privacy
Preserving



Upload Valid & Permitted Clips Only

Light-frontend;
Save storage & bandwidth

2. Offline Quality Check & Scene Labeling

2.1 Image-based Detection



Cloud Storage



Video Quality

Aspect,
integrity, exposure,
duration, decode



Valid Recording

Remove
hand-less clips &
non-egocentric &
Video Jitter



Security Info Erase

Privacy and
security info detect
& Erase



Hand Re-check

Third-vision &
truncation &
occlusion Ratio



Calibration

Camera intrinsics
check



Video Preprocessing

Slicing, frame-rate standardization
output video + metadata
(Lineage & Labels)



Compliance Detection

Semantic-level
re-check for
collection specs



Valid Action Detection

Semantic-level valid &
complete
manipulation
(VLM level)



Level-4 Label Tree Build

Domain, scene,
task, objects;

3. Reconstruction & Annotation



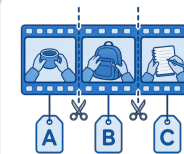
Camera Trajectory

SLAM estimates 6-DoF camera poses



Hand Reconstruction

Reconstruction hand 3D MANO
meshes, extract to 21-joint



Atomic Action Slicing & Labeling

Slice into atomic clips & label

4. Quality Inspection & Data Delivery

Three-Gate Quality Check



Completeness

Sufficient valid
frame ratio



Correctness

Low IK
failure rate



Consistency

Smooth & continuous
trajectories

Human Inspection



Re-check
sample data



Final Output Data & Training Corpus

~2,000 hours for
open-source release